Lab: Personal Expense Tracker

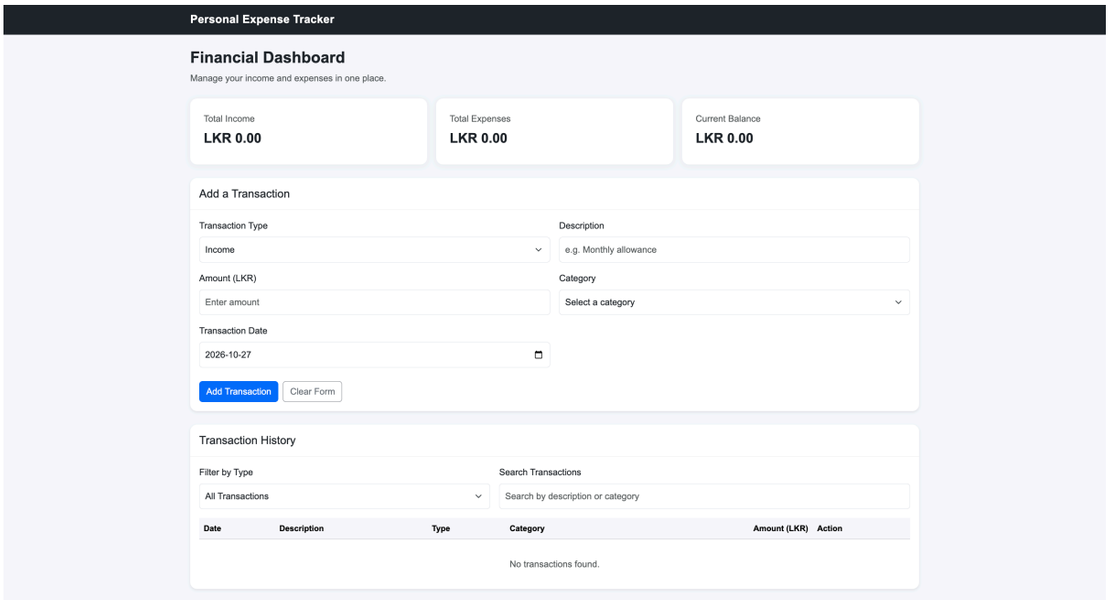
You have been asked to develop a Personal Expense Tracker for university students. The application should allow users to record their income and expenses, monitor their financial balance, and review their transaction history.

The application must work entirely on the client side. No backend server or database is required.

Task 1: Create the user interface

Create a responsive web application with the following components:

1. A navigation bar displaying the application name, Personal Expense Tracker.
2. Three summary cards showing:
    - Total Income
    - Total Expenses
    - Current Balance
3. A transaction form containing:
    - Transaction type: Income or Expense
    - Description
    - Amount
    - Category
    - Date
4. A transaction history table containing the date, description, type, category, amount, and action buttons.
5. A filter that allows users to view all transactions, income only, or expenses only.
6. A clear button to reset the form.



Task 2: Implement the functionality

Implement the following features using JavaScript.

Requirement	Description
Add transaction	Add a valid income or expense to the transaction list.
Form validation	Ensure that required fields are completed and the amount is greater than zero.
Calculate totals	Calculate total income, total expenses, and the current balance.
Dynamic table	Update the transaction table whenever a transaction is added or deleted.
Filter transactions	Display all transactions or only income or expenses.
Delete transaction	Allow users to delete an existing transaction.
Persistent storage	Save transactions in `localStorage` and restore them after refreshing the browser.

Responsive design	Ensure that the interface works on desktop and mobile screens.

Task 3: Test the application

Use the following data to test your application.

Type	Description	Category	Amount (LKR)
Income	Monthly Allowance	Allowance	25,000
Expense	Lunch	Food	800
Expense	Bus Fare	Transport	250
Income	Part-time Work	Salary	5,000

Expected results after entering all four transactions:

Income

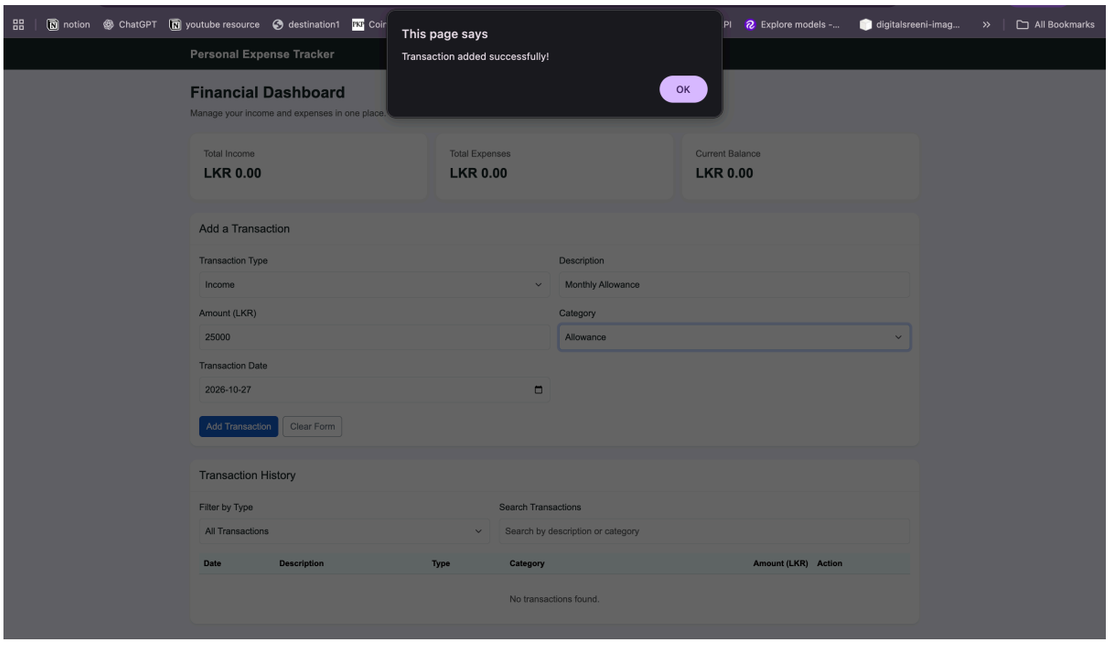
LKR 30,000

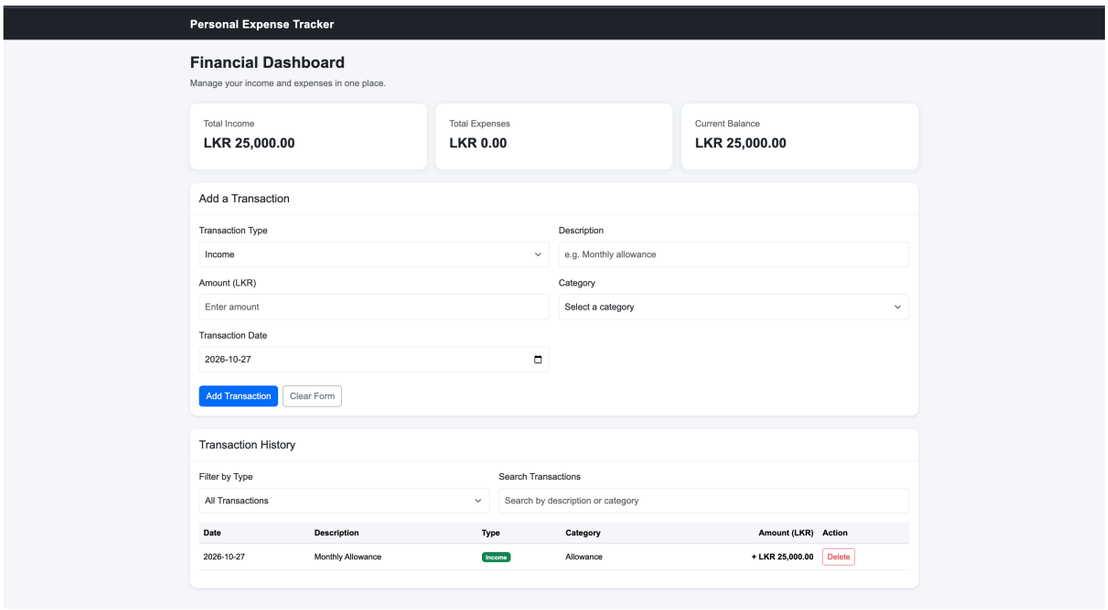
Expenses

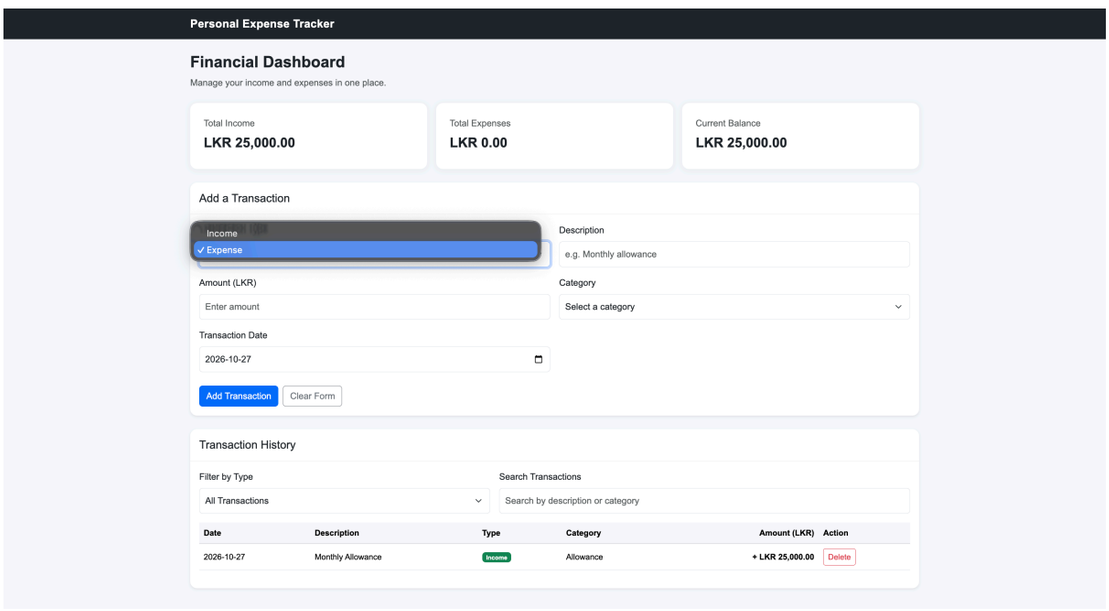
LKR 1,050

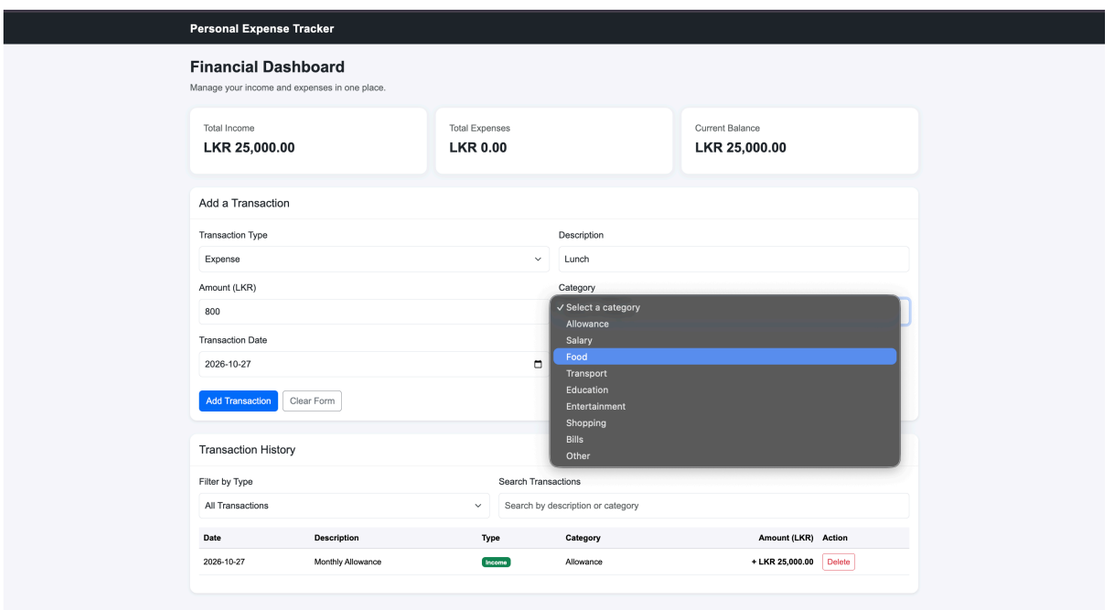
Balance

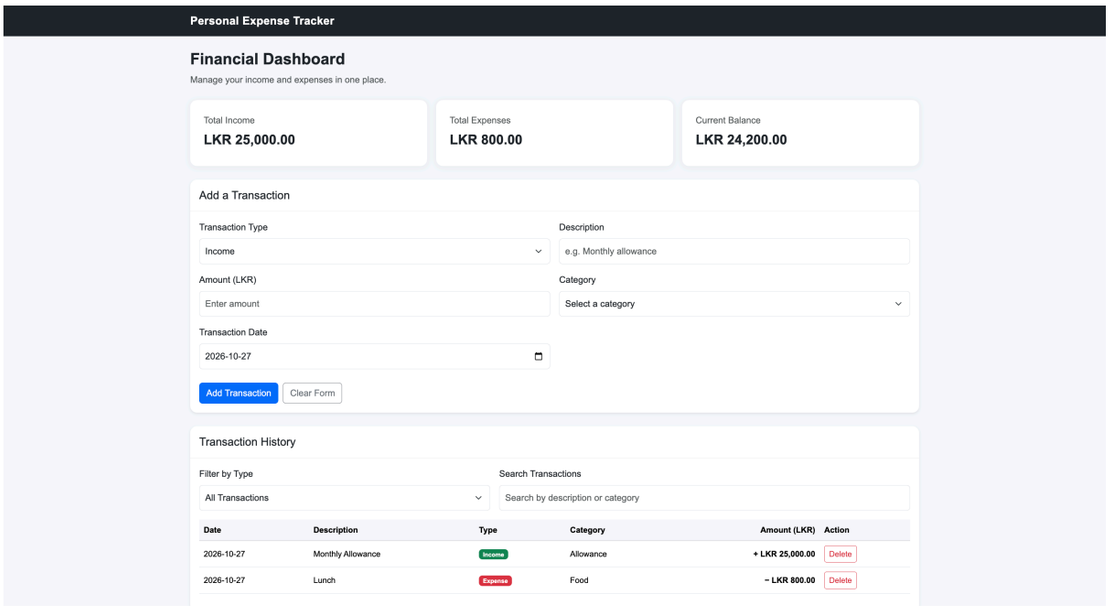
LKR 28,950

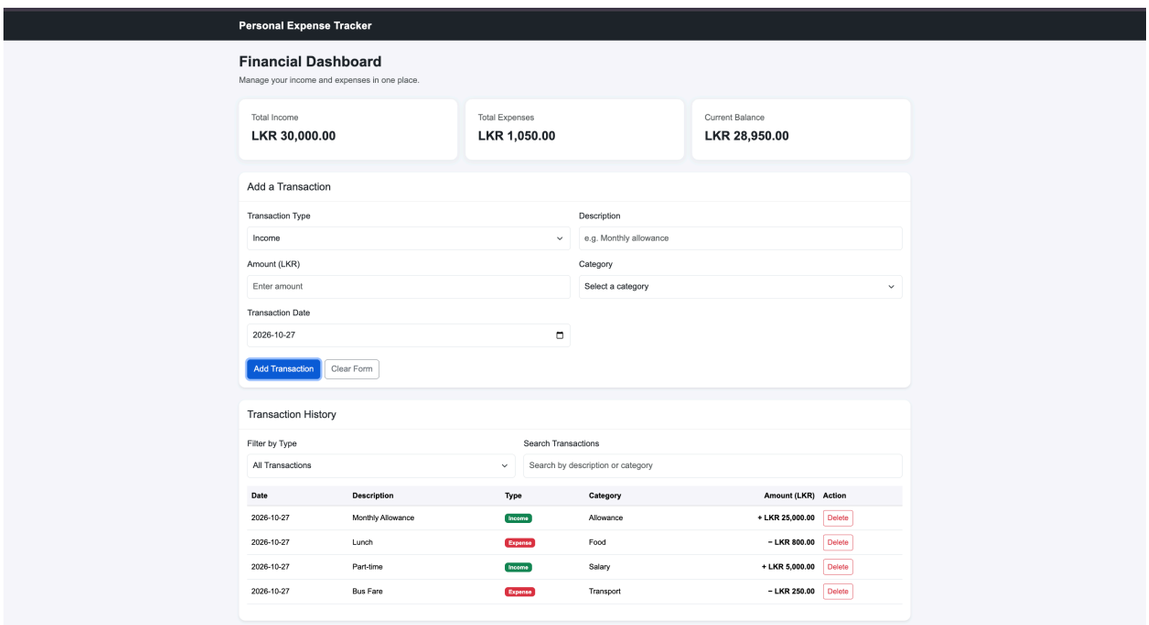


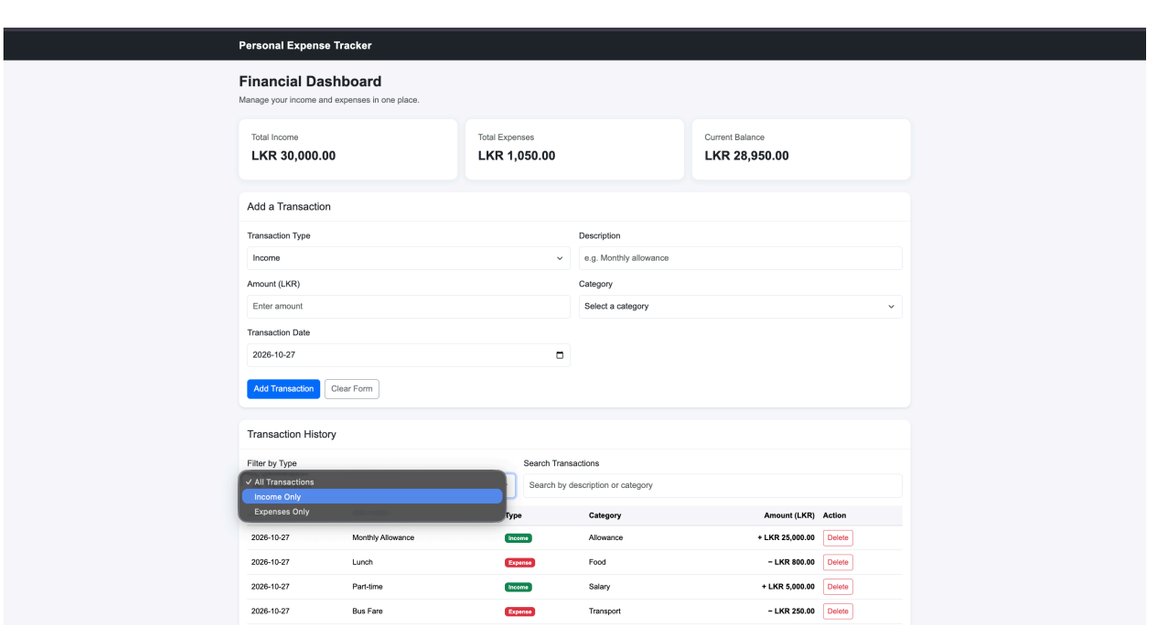

Personal Expense Tracker
Financial Dashboard
Manage your income and expenses in one place.
Total Income
LKR 25,000.00
Total Expenses
LKR 0.00
Current Balance
LKR 25,000.00
Add a Transaction
Transaction Type
Income
Description
e.g. Monthly allowance
Amount (LKR)
Enter amount
Category
Select a category
Transaction Date
2026-10-27
Add Transaction
Clear Form
Transaction History
Filter by Type
All Transactions
Search Transactions
Search by description or category
Date
Description
Type
Category
Amount (LKR)
Action
2026-10-27
Monthly Allowance
Income
Allowance
+ LKR 25,000.00
Delete

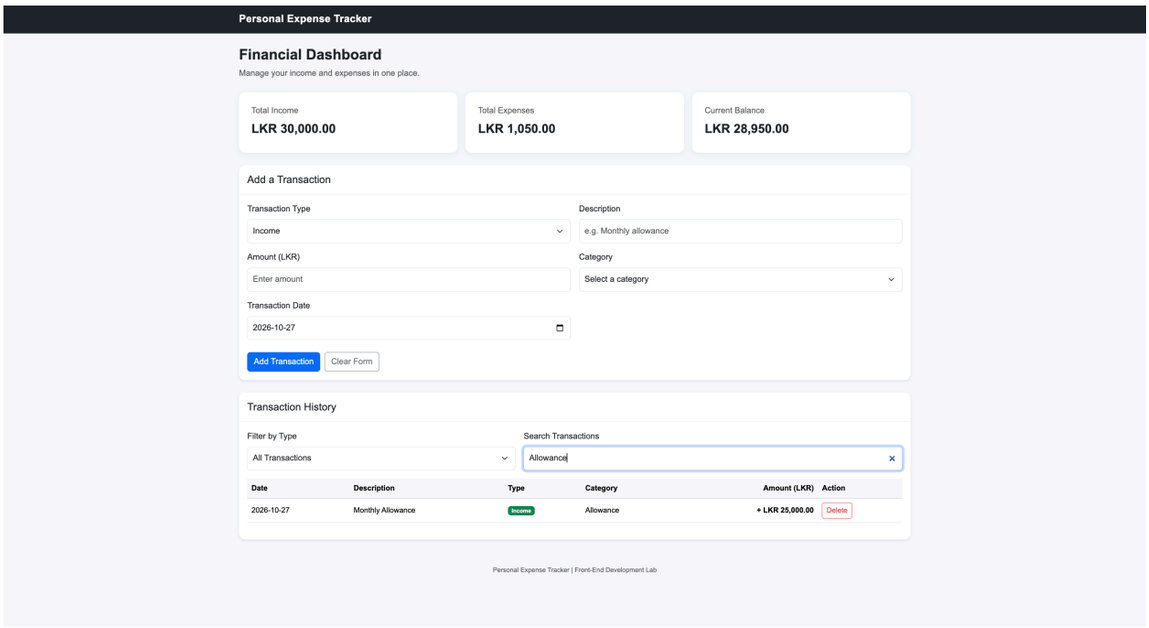

Personal Expense Tracker
Financial Dashboard
Manage your income and expenses in one place.
Total Income
LKR 25,000.00
Total Expenses
LKR 0.00
Current Balance
LKR 25,000.00
Add a Transaction
Income
Expense
Description
e.g. Monthly allowance
Amount (LKR)
Enter amount
Category
Select a category
Transaction Date
2026-10-27
Add Transaction
Clear Form
Transaction History
Filter by Type
All Transactions
Search Transactions
Search by description or category
Date
Description
Type
Category
Amount (LKR)
Action
2026-10-27
Monthly Allowance
Income
Allowance
+ LKR 25,000.00
Delete


Personal Expense Tracker
Financial Dashboard
Manage your income and expenses in one place.
Total Income
LKR 25,000.00
Total Expenses
LKR 0.00
Current Balance
LKR 25,000.00
Add a Transaction
Transaction Type
Expense
Description
Lunch
Amount (LKR)
800
Category
Select a category
Allowance
Salary
Food
Transport
Education
Entertainment
Shopping
Bills
Other
Transaction Date
2026-10-27
Add Transaction
Clear Form
Transaction History
Filter by Type
All Transactions
Search Transactions
Search by description or category
Date
Description
Type
Category
Amount (LKR)
Action
2026-10-27
Monthly Allowance
Income
Allowance
+ LKR 25,000.00
Delete


Personal Expense Tracker
Financial Dashboard
Manage your income and expenses in one place.
Total Income
LKR 25,000.00
Total Expenses
LKR 800.00
Current Balance
LKR 24,200.00
Add a Transaction
Transaction Type
Income
Description
e.g. Monthly allowance
Amount (LKR)
Enter amount
Category
Select a category
Transaction Date
2026-10-27
Add Transaction
Clear Form
Transaction History
Filter by Type
All Transactions
Search Transactions
Search by description or category
Date
Description
Type
Category
Amount (LKR)
Action
2026-10-27
Monthly Allowance
Income
Allowance
+ LKR 25,000.00
Delete
2026-10-27
Lunch
Expense
Food
− LKR 800.00
Delete


Personal Expense Tracker
Financial Dashboard
Manage your income and expenses in one place.
Total Income
LKR 30,000.00
Total Expenses
LKR 1,050.00
Current Balance
LKR 28,950.00
Add a Transaction
Transaction Type
Income
Description
e.g. Monthly allowance
Amount (LKR)
Enter amount
Category
Select a category
Transaction Date
2026-10-27
Add Transaction
Clear Form
Transaction History
Filter by Type
All Transactions
Search Transactions
Search by description or category
Date
Description
Type
Category
Amount (LKR)
Action
2026-10-27
Monthly Allowance
Income
Allowance
+ LKR 25,000.00
Delete
2026-10-27
Lunch
Expense
Food
− LKR 800.00
Delete
2026-10-27
Part-time
Income
Salary
+ LKR 5,000.00
Delete
2026-10-27
Bus Fare
Expense
Transport
− LKR 250.00
Delete


Personal Expense Tracker
Financial Dashboard
Manage your income and expenses in one place.
Total Income
LKR 30,000.00
Total Expenses
LKR 1,050.00
Current Balance
LKR 28,950.00
Add a Transaction
Transaction Type
Income
Description
e.g. Monthly allowance
Amount (LKR)
Enter amount
Category
Select a category
Transaction Date
2026-10-27
Add Transaction
Clear Form
Transaction History
Filter by Type
All Transactions
Income Only
Expenses Only
Search Transactions
Search by description or category
Type
Category
Amount (LKR)
Action
2026-10-27
Monthly Allowance
Income
Allowance
+ LKR 25,000.00
Delete
2026-10-27
Lunch
Expense
Food
− LKR 800.00
Delete
2026-10-27
Part-time
Income
Salary
+ LKR 5,000.00
Delete
2026-10-27
Bus Fare
Expense
Transport
− LKR 250.00
Delete



Task 4: Host the Personal Expense Tracker

Deploy the completed Personal Expense Tracker using a freely available hosting service.

Instructions

1. Create a GitHub repository named `personal-expense-tracker`.
2. Upload `index.html`, `style.css`, and `script.js` to the repository.
3. Host the application using one of the following services:
    - **GitHub Pages:** https://pages.github.com/
    - **Netlify:** https://www.netlify.com/
    - **Vercel:** https://vercel.com/
4. Verify that the hosted application works correctly in a web browser.
5. Test adding, deleting, filtering, and searching transactions.
6. Refresh the page and verify that previously saved transactions are still available.
7. Submit both the GitHub repository URL and the deployed application URL.

Submission format

- **GitHub Repository URL:** _____
- **Hosted Application URL:** _____

Note: The application must be publicly accessible through the submitted URL. Ensure that all HTML, CSS, and JavaScript files are included and that the application functions correctly after deployment.